

CS218 DAA Programming Assignment 1

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Algorithm:

1. Create a Minimum Spanning Tree by adding lowest weight edges for vertices that are not yet connected, by giving preference first to blue edges and then red edges. $O(E \log E)$
2. If weight less than threshold, return. Else, choose a red edge to swap with blue edge in order to reduce weight. A blue edge can't be swapped with another blue edge because the smallest weighted edges are already selected.
3. Iterate over all red edges not in Spanning Tree, add them to ST and pick a blue edge from the cycle created such that swapping them maximally reduces total weight. Add these pairs to heap valued by reduced weight. $O(V_r(V + E))$
4. Loop till weight $\leq$ threshold and swap edges from the edge pair on the top of the heap. Modify the heap accordingly after each successful/failed swap. If cycle is detected after swap (precomputed values no more work), revert and check again for better match. $O(V + E)$

Compilation instructions (for Linux)

```
g++ 23b1016.cpp -fsanitize=address -o 23b1016.out  
Run 23b1016.out
```